SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Rising Rookies

Selected Domain

Domain 1 - Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

4 - Quality Education (primary); 8, 9, 13, 16 supporting

Project / Solution Name

PathFinder

Name (Leader)

LIN ZIRUI

Email & Phone (Leader)

1094184601@qq.com | Phone: PENDING

University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

University and TVET learners, fresh graduates and career switchers face a practical decision: which skills should they learn next as AI changes work? A self-reported skills list alone does not reveal relevant occupations, missing foundations or an affordable sequence of learning activities. A static course list also hides the evidence behind its recommendations.

PathFinder focuses on this decision for learners exploring technical careers, starting with five curated role pathways. It combines a synthetic skills profile with dated public job and AI-discussion signals, explains the score and gaps, and converts the selected role into an introductory learning plan. The Domain 1 brief identifies a Malaysian reskilling need [D1, pp. 1-3]. The prototype uses an international, Europe-heavy job sample and makes no claim to measure Malaysian hiring demand.

The intended social benefit is access to understandable skills guidance and low-cost learning, aligned with SDG 4.4 and 4.3. Employment, income and environmental outcomes have not been measured. No real CV, personal learner record or hiring decision enters the demonstration.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

PathFinder proposes an evidence-grounded career recommendation system. The deployed prototype exposes a deterministic baseline: keyword extraction, separate job and AI-discussion channels, weighted role matching and a curated learning catalogue. No trained forecasting model or language model runs in the application.

The six Domain 1 capabilities are implemented as a synthetic profile form; two refreshable public feeds with dates and links; five role pathways with score components and gaps; ordered introductory activities with estimated hours; role and skill links to SDG 4 and application-related goals; and moderate versus fast scenarios with explicit assumptions. Within-source directions require at least five records in each date group.

A planned AI extension would classify ambiguous skills with a declared model constrained to verified references and abstain on unsupported content. It is not deployed. The team must decide whether the baseline sufficiently meets the domain's AI-powered expectation.

Cloud hosting and data validation work today. Replit execution support is prepared, but actual team-account use and workshop participation remain unverified. No sponsor GenAI integration is claimed. Major AI-assisted development is disclosed in the Technical Tools Declaration. Registered members must review, adapt and explain the source and proposal before submission.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Implemented: HTML5, CSS3 and browser JavaScript; a separate testable recommendation module; Fetch API for Arbeitnow and HN Search by Algolia; browser localStorage for synthetic selections and progress; JSON import/export; static hosting through ChatGPT Sites on Cloudflare. There is no application database, profile backend, runtime LLM or GPU workload. Python 3 standard-library HTTP serving and a .replit configuration support a future team-account Replit run.

Data flow: synthetic profile + dated public metadata → validation and deduplication → source-specific keyword counts → role scoring and gaps → prerequisite-ordered introductory plan → SDG explanations and local export. This is a logical pipeline, not a trained AI model.

Sponsor resources actually used in the observed workflow: none evidenced. Replit is compulsory under the orientation rules, so use and evidence from the team's own account remain a submission blocker. RunPod is not used because no GPU work runs; the sample README's unconditional wording needs organizer clarification. See the attached proof/status PDFs. Planned model/API services must be added to the declaration only after actual use.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The design starts from a learner's task: choose a plausible direction and one next learning step. A synthetic persona, skill checkboxes and weekly-hours input reduce setup effort. Separate views expose evidence, scoring, learning milestones and SDG rationale. The approach is a desk-based design hypothesis; no user interviews or usability study have been completed or claimed.

The working baseline supports manual feed refresh and dated JSON import when feeds fail. It retains prior evidence on total refresh failure and does not invent records. An update recalculates scores while preserving the learner's selected target. Core checks cover evidence effects, scenario differences, gap changes, source separation, malformed imports, study ordering and score bounds.

Before Phase 2, the registered team should review every module, run the build in its own Replit account, verify resource rights, test five synthetic journeys and measure completion time. A small usability pilot can then test comprehension and accessibility. AI integration, Malaysian datasets, database-backed history and automatic monitoring remain future work. Deployments and genuine commits must stop before the preliminary deadline.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Primary track: SDG 4. The baseline below is a synthetic evaluation design, not a measured Malaysian population statistic. References: [D1] Domain 1 brief; [UN4] https://sdgs.un.org/goals/goal4. No carbon, salary or job-placement effect is claimed.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 4.4, relevant technical and vocational skills for work; SDG 4.3, affordable access to learning. Supporting: SDG 8.2 and 8.5, productive work and career transitions. Role-specific applications also connect to 9, 13 and 16. Source: UN SDG goal/target pages [UN4, UN8]. Contributions are plausible pathways, not verified outcomes.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Design baseline: 0 of 5 specified synthetic cases have an exported, source-linked plan when the baseline is a profile-only form (0%). This is an explicit constructed comparator, not a user study. Human planning time and labour-market outcomes: not measured. The team must record the same five cases, inputs, outputs and test date before reporting achieved improvement.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: completed synthetic journeys / 5 × 100%. A journey passes only if it has a score breakdown, explicit gap list, ordered plan, time estimate, evidence links and SDG links. Quality checks: 100% of displayed role/skill recommendations have catalogue references and SDG reasons. Record feed-retrieval date and failure behaviour.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 target: at least 4/5 (80%) synthetic journeys pass the defined acceptance checklist. From a 0% constructed comparator, this is an 80 percentage-point target. Usability target: at least 80% of pilot participants finish within 5 minutes; not yet tested. No employment, income or environmental gain is forecast as an achieved result.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

CHECKLIST STATUS
Workshop completion: PENDING confirmation for all five members.
Problem/SDG alignment and working UI: documented in this draft.
All required fields complete: NO; phone, evidence and signed team declarations remain pending.
Boxes are intentionally not certified on the team's behalf.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

PENDING TEAM CONFIRMATION. No Yes/No selection is asserted. The visible development of this prototype occurred on 23 October 2026, after the orientation release date of 19 October. This does not establish whether any member contributed earlier code, research or design. Each member must disclose prior-owned material and ownership, or confirm None, before the leader signs. Major AI-assisted creation is separately disclosed; do not present these files as independently written team work.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Resources (YES)	Type	Licence / terms
HTML/CSS/JS browser APIs	Web platform; original app code	Standards; project code Apache 2.0 subject to team IP confirmation
Arbeitnow API; HN Search	Public metadata feeds	Provider/publisher terms; no unrestricted data licence asserted
O*NET® OnLine	Role reference links	USDOL/ETA; CC BY 4.0, with project-specific adaptations noted
UN SDG and learning sites	References and course links	Provider terms; linked, not copied
OpenAI ChatGPT / Codex	AI-assisted development, major	OpenAI terms; exact model ID not exposed

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

PENDING TEAM CONFIRMATION. No Yes/No selection is asserted. The five registered members must check employment, internship and consulting obligations and identify affected work or confirm None. The assistant cannot determine ownership or sign a personal declaration for the team.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

PENDING: LIN ZIRUI must sign only after reviewing and adapting the materials, completing all required evidence, and confirming team originality and IP status.

[Date]

PENDING actual signing date. Draft prepared: 23 October 2026.

NOT READY FOR SUBMISSION
Phone, GitHub, Replit proof, workshop confirmation, prior-work / employment-IP answers and a genuine signature remain unresolved. AI disclosure alone does not establish compliance with the originality rule.

Requirements and verification appendix

Evidence status: review draft; unresolved fields are not certified.

Implemented scope

Profile form with fixed synthetic personas; 9 skills; 5 role pathways; manual updates from 2 public feeds; import validation and source separation; score breakdowns and gaps; ordered introductory resources; scenario comparison; SDG mappings; local milestones and JSON export. No CV upload, runtime LLM, background monitoring, real recruitment or job guarantee.

Acceptance and evidence

Core test command: node tests/core.test.cjs. Tests verify baseline scores, evidence effects, scenario differences, skill-gap updates, score bounds, source isolation, low-sample guards, keyword boundaries, invalid URL/date rejection, deduplication, learning order, duration and exclusion of demographic fields. Passing these checks is software verification, not a user-impact study.

Primary references

[D1] Domain 1(1).pdf, internal pages 1-7 (PDF pages 2-8): problem, six features, rules and preliminary rubric. [O] SDC Hackathon Online Orientation Slides, PDF pp. 21-24 and 28-35: domain, AI/integrity, submissions, sponsor resources, deadline and penalties. [P] Official Phase 1 Submission Template. [T] Official Technical Tools Declaration Template.

SDG and provider references

[UN4] https://sdgs.un.org/goals/goal4; [UN8] https://sdgs.un.org/goals/goal8. [J] https://www.arbeitnow.com/blog/job-board-api. [HN] https://hn.algolia.com/api. [ON] https://www.onetonline.org/help/license. Access checked 23 October 2026; external page access can change.

Requirements and verification appendix

Evidence status: review draft; unresolved fields are not certified.

Word-limit audit

Proposal 2.1: 166/250 words. Proposal 2.2: 191/300. Proposal 3.2: 171/200. Declaration 2.1: 134/200. AI review: 79/100. Privacy: 121/150. Risks: 109/150. Counts use whitespace-separated English words.

Evaluation design and targets

Five synthetic journeys should cover the different target pathways and varied skill coverage. Record fixture ID, inputs, scenario, source snapshot, target, gap list, references, total hours and export. Baseline 0/5 is a constructed profile-only comparator. Phase 2 target >=4/5 (80%) passing journeys is proposed, not achieved. Human planning-time baseline and outcome data remain unmeasured.

Completion blockers

PENDING: leader phone; actual team public GitHub URL; actual team-account Replit usage/proof; five members’ workshop completion; prior-work and employment-IP answers; source rights review; team understanding/adaptation of AI output; genuine leader signature/date. Runtime AI functionality is not implemented; the team should confirm whether the deterministic baseline is adequate for the domain.

Official GitHub sample

Retrieved directly from https://github.com/michaelzhngg/SDC_Hackathon_Submission_Template using the GitHub contents/tree API on 23 October 2026. Its sample includes preliminary, technology, Replit proof and RunPod proof PDFs. The README marks RunPod required; orientation p.29 limits the RunPod rule to GPU work. Obtain written organizer clarification. Terms & Conditions take precedence; they were not provided.